

SOLIDWORKS API (.NET, VBA, C#) — Reference (276/337): ISketchPoint-ISketchRegion

SOLIDWORKS API Documentation (converted from CHM)

ISketchPoint (Type)

ISketchPoint Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchPoint Interface

Provides access to sketch point entities.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchPoint
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint
```

C#

```
public interface ISketchPoint
```

C++/CLI

```
public interface class ISketchPoint
```

Visual Basic for Applications (VBA) Syntax

See [SketchPoint](#).

Example

[Get Sketch Points and Sketch Point IDs \(VBA\)](#)
[Get Sketch Points \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

The ISketchPoint object provides access to sketch points while the [ISketchSegment](#) object provides access to sketch arcs, lines, ellipses, parabolas, and splines and their corresponding [ISketchArc](#), [ISketchEllipse](#), [ISketchLine](#), [ISketchParabola](#), [ISketchText](#), and [ISketchSpline](#) interfaces.

Accessors

[IChainPatternFeatureData::Group1PathLink1](#)
[IChainPatternFeatureData::Group1PathLink2](#)
[IChainPatternFeatureData::Group2PathLink1](#)
[IChainPatternFeatureData::Group2PathLink2](#)
[ICoincidentMateFeatureData::EntitiesToMate](#)
[IConcentricMateFeatureData::EntitiesToMate](#)
[IDomeFeatureData2::ConstraintPointOrSketch](#)
[IEnumSketchPoints::Next](#)
[IFillPatternFeatureData::SeedFeatureCenter](#)
[IHingeMateFeatureData::EntitiesToMate](#)
[IHoleSeriesFeatureData2::GetSketchPoints](#) and [IHoleSeriesFeatureData2::!GetSketchPoints](#)
[ILayer::GetItems](#)

[ILocalSketchPatternFeatureData::SelectedPoint](#)
[IModelDoc2::CreatePoint2](#) and [IModelDoc2::ICreatePoint2](#)
[IMoveCopyBodyFeatureData::TransformReferenceEntity](#)
[IMoveCopyBodyFeatureData::TranslateToVertex](#)
[IPartialEdgeFilletData::ReferenceOffsetEnd](#)
[IPartialEdgeFilletData::ReferenceOffsetStart](#)
[IPrimaryMemberPointLengthFeatureData::EndPoint](#)
[IPrimaryMemberPointLengthFeatureData::GetPoints](#)
[ISecondaryMemberUpToMembersFeatureData::GetFromPoint](#)
[ISecondaryMemberUpToMembersFeatureData::GetPointMemberPairs](#)
[ISketch::GetSketchPoints2](#)
[ISketchArc::GetCenterPoint2](#) and [ISketchArc::IGetCenterPoint2](#)
[ISketchArc::GetEndPoint2](#) and [ISketchArc::IGetEndPoint2](#)
[ISketchArc::GetStartPoint2](#) and [ISketchArc::IGetStartPoint2](#)
[ISketchEllipse::GetCenterPoint2](#) and [ISketchEllipse::IGetCenterPoint2](#)
[ISketchEllipse::GetEndPoint2](#) and [ISketchEllipse::IGetEndPoint2](#)
[ISketchEllipse::GetMajorPoint2](#) and [ISketchEllipse::IGetMajorPoint2](#)
[ISketchEllipse::GetMinorPoint2](#) and [ISketchEllipse::IGetMinorPoint2](#)
[ISketchEllipse::GetStartPoint2](#) and [ISketchEllipse::IGetStartPoint2](#)
[ISketchLine::GetEndPoint2](#) and [ISketchLine::IGetEndPoint2](#)
[ISketchLine::GetStartPoint2](#) and [ISketchLine::IGetStartPoint2](#)
[ISketchParabola::GetApexPoint2](#) and [ISketchParabola::IGetApexPoint2](#)
[ISketchParabola::GetEndPoint2](#) and [ISketchParabola::IGetEndPoint2](#)
[ISketchParabola::GetFocalPoint2](#) and [ISketchParabola::IGetFocalPoint2](#)
[ISketchParabola::GetStartPoint2](#) and [ISketchParabola::IGetStartPoint2](#)
[ISketchPatternFeatureData::ReferencePoint](#)
[ISketchSlot::GetCenterPointHandle](#)
[ISketchSlot::GetSketchPoints](#)
[ISMGussetFeatureData::ReferencePoint](#)
[IStructuralMemberFeatureData::GetConnectionPoints](#) and [IStructuralMemberFeatureData::IGetConnectionPoints](#)
[IStructuralMemberFeatureData::LocateProfilePoint](#)
[IStructuralMemberGroup::LocateProfilePoint](#)
[IStructureSystemMemberProfile::PiercePointSelectionObject](#)
[ISurfaceFlattenFeatureData::FixPointVertex](#)
[ISymmetricMateFeatureData::EntitiesToMate](#)
[ITabAndSlotGroupData::SelectionEndReferencePoint](#)
[ITabAndSlotGroupData::SelectionStartReferencePoint](#)
[ITablePatternFeatureData::ReferencePoint](#)
[IUniversalJointMateFeatureData::JointPoint](#)
[IWizardHoleFeatureData2::GetSketchPoints](#) and [IWizardHoleFeatureData2::IGetSketchPoints](#)
[IWrapSketchFeatureData::PullDirection](#)

Access Diagram

[SketchPoint](#)

See Also

[ISketchPoint Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketch::MergePoints Method](#)
[IEnumSketchPoints Interface](#)
[IRefPoint Interface](#)

ISketchPoint Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISketchPoint Interface

For a list of all members of this type, see [ISketchPoint members](#).

Public Methods

Name	Description
 DeSelect	Deselects the sketch point.
 GetCoords	Gets the x coordinate of the sketch point.
 GetFramePointTangent	Obsolete. Not superseded.
 GetID	Gets the sketch point ID for this sketch point.
 GetRelations	Gets the sketch relations for this sketch point.
 GetRelationsCount	Gets the number of sketch relations for this sketch point.
 GetSketch	Gets the sketch for the current sketch point.
 GetSketchSlot	Gets sketch slot with which this sketch point is associated.
 IGetFramePointTangent	Obsolete. Not superseded.
 IGetID	Gets the sketch point ID for this sketch point.
 IGetRelations	Gets the sketch relations for this sketch point.
 ISetFramePointTangent	Obsolete. Not superseded.
 Select	Obsolete. Superseded by ISketchPoint.Select4 .
 Select2	Obsolete. Superseded by ISketchPoint.Select4 .
 Select3	Obsolete. Superseded by ISketchPoint.Select4 .
 Select4	Selects the sketch point and marks it.
 SelectByMark	Obsolete. Superseded by ISketchPoint.Select4 .
 SetCoords	Sets the location of this sketch point.
 SetFramePointTangent	Obsolete. Not superseded.

[Top](#)

See Also

[ISketchPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketch::MergePoints Method](#)
[IEnumSketchPoints Interface](#)
[IRefPoint Interface](#)

ISketchPoint Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISketchPoint Interface

The following tables list the members exposed by [ISketchPoint](#).

Public Properties

Name	Description
 Color	Gets or sets the color of this sketch point.
 Layer	Gets or sets the layer for this sketch point.
 LayerOverride	Gets or sets whether the sketch point has properties that override the default properties of the layer.
 Status	Gets or sets the status of the associated relation for this sketch point.
 Type	Gets or sets the type of sketch point.

 X	Gets or sets the x coordinate of the sketch point.
 Y	Gets or sets the y coordinate of the sketch point.
 Z	Gets or sets the z coordinate of the sketch point.

[Top](#)

Public Methods

Name	Description
 DeSelect	Deselects the sketch point.
 GetCoords	Gets the x coordinate of the sketch point.
 GetFramePointTangent	Obsolete. Not superseded.
 GetID	Gets the sketch point ID for this sketch point.
 GetRelations	Gets the sketch relations for this sketch point.
 GetRelationsCount	Gets the number of sketch relations for this sketch point.
 GetSketch	Gets the sketch for the current sketch point.
 GetSketchSlot	Gets sketch slot with which this sketch point is associated.
 IGetFramePointTangent	Obsolete. Not superseded.
 IGetID	Gets the sketch point ID for this sketch point.
 IGetRelations	Gets the sketch relations for this sketch point.
 ISetFramePointTangent	Obsolete. Not superseded.
 Select	Obsolete. Superseded by ISketchPoint.Select4 .
 Select2	Obsolete. Superseded by ISketchPoint.Select4 .
 Select3	Obsolete. Superseded by ISketchPoint.Select4 .
 Select4	Selects the sketch point and marks it.
 SelectByMark	Obsolete. Superseded by ISketchPoint.Select4 .
 SetCoords	Sets the location of this sketch point.
 SetFramePointTangent	Obsolete. Not superseded.

[Top](#)

See Also

[ISketchPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketch.MergePoints Method](#)
[IEnumSketchPoints Interface](#)
[IRefPoint Interface](#)

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ISketchPoint Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchPoint Interface

For a list of all members of this type, see [ISketchPoint members](#).

Public Properties

Name	Description
 Color	Gets or sets the color of this sketch point.
 Layer	Gets or sets the layer for this sketch point.
 LayerOverride	Gets or sets whether the sketch point has properties that override the default properties of the layer.
 Status	Gets or sets the status of the associated relation for this sketch point.
 Type	Gets or sets the type of sketch point.

 X	Gets or sets the x coordinate of the sketch point.
 Y	Gets or sets the y coordinate of the sketch point.
 Z	Gets or sets the z coordinate of the sketch point.

[Top](#)

See Also

[ISketchPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[ISketchPoint.MergePoints Method](#)
[IEnumSketchPoints Interface](#)
[IRefPoint Interface](#)

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Color Property (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Color Property (ISketchPoint)

Gets or sets the color of this sketch point.

.NET Syntax

Visual Basic (Declaration)

Property Color As System.Integer

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim value As System.Integer
```

```
instance.Color = value
```

```
value = instance.Color
```

C#

```
System.int Color {get; set;}
```

C++/CLI

```
property System.int Color {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

COLORREF value for this sketch point

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.Color](#).

Remarks

Only drawing documents support layers.

Sketch points can be created on a layer that has specific visual properties. The color value specified with this property overrides the layer color.

Use the [ISketchPoint.LayerOverride](#) to determine if this sketch point is currently using its default layer color.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint.Layer Property](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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DeSelect Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : DeSelect Method (ISketchPoint)

Deselects the sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function DeSelect() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Boolean
```

```
value = instance.DeSelect()
```

C#

```
System.bool DeSelect()
```

C++/CLI

```
System.bool DeSelect();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the sketch point is deselected, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.DeSelect](#).

Remarks

This method does not work well when a PropertyManager page is open or a command is running. Use [IModelDoc2::DeSelectByID](#) instead of using this method. IModelDoc2::DeSelectByID handles selection correctly whether or not a command is running.

To select or deselect a sketch point, the owning document of that ISketchPoint object needs to be open and visible.

Sketch point selections are accessible through the [ISelectionMgr](#) of the owning document of the ISketchPoint object, even if the owning document is not active.

Selection or deselection does not work for a sketch point in a document within a drawing. Selection or deselection of sketch points owned by the drawing does work, but only when the drawing document is active.

If the owning sketch of a sketch point was active, or inactive, when the sketch point was obtained, then it must be active, or inactive, to deselect it. For example, if the owning sketch of a sketch point was active when the sketch point was obtained, then the owning sketch must be active to select or deselect the sketch point. Likewise, if the owning sketch of a sketch point was inactive when the sketch point was obtained, then the owning sketch must be inactive to select or deselect the sketch point

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::Select4 Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

GetCoords Method (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetCoords Method (ISketchPoint)

Gets the x coordinate of the sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetCoords() As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Double  
  
value = instance.GetCoords()
```

C#

```
System.double GetCoords()
```

C++/CLI

```
System.double GetCoords();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

x coordinate of the sketch point

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetCoords](#).

Example

[Get Sketch Points \(VBA\)](#)

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::X Property](#)
[ISketchPoint::Y Property](#)
[ISketchPoint::Z Property](#)
[ISketchPoint::SetCoords Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

GetFramePointTangent Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetFramePointTangent Method (ISketchPoint)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFramePointTangent( _  
    ByRef Status As System.Boolean _
```

```
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim Status As System.Boolean  
Dim value As System.Object  
  
value = instance.GetFramePointTangent(Status)
```

C#

```
System.object GetFramePointTangent(  
    out System.bool Status  
)
```

C++/CLI

```
System.Object^ GetFramePointTangent(  
& [Out] System.bool Status  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Status

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetFramePointTangent](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

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GetID Method (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetID Method (ISketchPoint)

Gets the sketch point ID for this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetID() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Object  
  
value = instance.GetID()
```

C#

```
System.object GetID()
```

C++/CLI

```
System.Object^ GetID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array with two longs or two integers (see [Long vs. Integer](#)) identifying this sketch point ID

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetID](#).

Example

[Get Sketch Points and Sketch Point IDs \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Get Persistent Identifiers and Types for Sketch Points \(VB.NET\)](#)
[Get Persistent Identifiers and Types for Sketch Points \(C#\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

Remarks

The ID of the sketch point:

- is an ordered pair (i1, i2). For sketch points, the combination of these two numbers is always unique within a specific sketch.
- cannot be assigned by applications or users.
- is not the same as a [persistent reference ID](#).

Each point within a specific sketch has a unique ID. However, a point and other sketch objects can have the same ID. Likewise, in a second sketch, you may find a different sketch element with the same ID. Therefore, your application must keep track of:

- sketch element type (that is, point, line, arc, spline, and so on)
- owning sketch name
- sketch element ID to uniquely identify a sketched item

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetID Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

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GetRelations Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetRelations Method (ISketchPoint)

Gets the sketch relations for this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelations() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Object
```

```
value = instance.GetRelations()
```

C#

```
System.object GetRelations()
```

C++/CLI

```
System.Object^ GetRelations();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch relations](#)

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetRelations](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetRelationsCount Method](#)
[ISketchPoint::IGetRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetRelationsCount Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetRelationsCount Method (ISketchPoint)

Gets the number of sketch relations for this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetRelationsCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Integer  
  
value = instance.GetRelationsCount()
```

C#

```
System.int GetRelationsCount()
```

C++/CLI

```
System.int GetRelationsCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of sketch relations for this sketch point

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetRelationsCount](#).

Remarks

Call this method before calling [ISketchPoint::IGetRelations](#) to get the size of the array for that method.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetRelations Method](#)

Availability

GetSketch Method (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetSketch Method (ISketchPoint)

Gets the sketch for the current sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function GetSketch() As Sketch
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As Sketch
```

```
value = instance.GetSketch()
```

C#

```
Sketch GetSketch()
```

C++/CLI

```
Sketch^ GetSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch](#)

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.GetSketch](#).

Example

[Get Sketch Point's View \(VBA\)](#)

[Get x,y,z Locations of Points in Sketch \(VBA\)](#)

[Redirect Spotlight \(VBA\)](#)

Remarks

You can use the [ISketch](#) interface for extracting information about geometry in the sketch.

See Also

[ISketchPoint Interface](#)

[ISketchPoint Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

GetSketchSlot Method (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : GetSketchSlot Method (ISketchPoint)

Gets sketch slot with which this sketch point is associated.

.NET Syntax

```
Function GetSketchSlot() As SketchSlot
```



```
Dim instance As ISketchPoint
Dim value As SketchSlot

value = instance.GetSketchSlot()
```

```
SketchSlot GetSketchSlot()
```

```
SketchSlot^ GetSketchSlot();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Sketch slot](#) if the sketch point is associated with a sketch slot, or null if the sketch point is not associated with a sketch slot

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.GetSketchSlot](#).

Example

[Get Sketch Slot Using Sketch Point and Segment \(C#\)](#)
[Get Sketch Slot Using Sketch Point and Segment \(VB.NET\)](#)
[Get Sketch Slot Using Sketch Point and Segment \(VBA\)](#)

Remarks

Sketch slot information is independent of the sketch point type.

A sketch slot is returned if it is associated with the sketch point regardless if the sketch point is an internal point or a user point.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

Availability

SOLIDWORKS 2009 SP2, Revision Number 17.2

IGetFramePointTangent Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : IGetFramePointTangent Method (ISketchPoint)

Obsolete. Not superseded.

.NET Syntax

```
Function IGetFramePointTangent( _
    ByRef Status As System.Boolean _
) As System.Double
```



```
Dim instance As ISketchPoint
Dim Status As System.Boolean
Dim value As System.Double

value = instance.IGetFramePointTangent(Status)
```

C#

```
System.double IGetFramePointTangent(
    out System.bool Status
)
```

C++/CLI

```
System.double IGetFramePointTangent(
    & [Out] System.bool Status
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Status

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.IGetFramePointTangent](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

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IGetID Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : IGetID Method (ISketchPoint)

Gets the sketch point ID for this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetID() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim value As System.Integer
```

```
value = instance.IGetID()
```

C#

```
System.int IGetID()
```

C++/CLI

```
System.int IGetID();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array with two longs or integers (see [Long vs. Integer](#)) identifying this sketch point ID

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::GetID](#).

Remarks

The ID of the sketch point:

- is an ordered pair (i1, i2). For sketch points, the combination of these two numbers is always unique within a specific sketch.
- cannot be assigned by applications or users.
- is not the same as a [persistent reference ID](#).

Each point within a specific sketch has a unique ID. However, a point and other sketch objects can have the same ID. Likewise, in a second sketch, you may find a different sketch element with the same ID. Therefore, your application must keep track of:

- sketch element type (that is, point, line, arc, spline, and so on)
- owning sketch name
- sketch element ID to uniquely identify a sketched item

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetID Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

IGetRelations Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : IGetRelations Method (ISketchPoint)

Gets the sketch relations for this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetRelations( _  
    ByVal Count As System.Integer _  
) As SketchRelation
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim Count As System.Integer  
Dim value As SketchRelation  
  
value = instance.IGetRelations(Count)
```

C#

```
SketchRelation IGetRelations(  
    System.int Count  
)
```

C++/CLI

```
SketchRelation^ IGetRelations(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count

Number of sketch relations

Return Value

- in-process, unmanaged C++: Pointer to an array of [sketch relations](#)

- VBA, VB.NET, C#, and C++/CLI: Not supported
See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketchPoint::GetRelationsCount](#) to get the value for Count.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetRelations Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ISetFramePointTangent Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : ISetFramePointTangent Method (ISketchPoint)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function ISetFramePointTangent(  
    ByRef ToVector As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim ToVector As System.Double  
Dim value As System.Boolean  
  
value = instance.ISetFramePointTangent(ToVector)
```

C#

```
System.bool ISetFramePointTangent(  
    ref System.double ToVector  
)
```

C++/CLI

```
System.bool ISetFramePointTangent(  
& System.double% ToVector  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ToVector

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::ISetFramePointTangent](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

Layer Property (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Layer Property (ISketchPoint)

Gets or sets the layer for this sketch point.

.NET Syntax

Visual Basic (Declaration)

Property Layer As System.String

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.String
```

```
instance.Layer = value
```

```
value = instance.Layer
```

C#

```
System.string Layer {get; set;}
```

C++/CLI

```
property System.String^ Layer {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Name of the layer for this sketch point

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.Layer](#).

Remarks

Layers are only supported in drawing documents.

Every sketch point is on a layer. Sketch points can be created on a specific layer in drawings.

The return value may be an empty string because an older document may not contain layers. Also, this can occur if sketch points were generated in a new document that has no layers defined.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint.LayerOverride Property](#)

Availability

SOLIDWORKS 99, datecode 1999207

LayerOverride Property (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : LayerOverride Property (ISketchPoint)

Gets or sets whether the sketch point has properties that override the default properties of the layer.

Visual Basic Syntax

Visual Basic (Declaration)

Property LayerOverride As System.Integer

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim value As System.Integer

instance.LayerOverride = value

value = instance.LayerOverride
```

C#

```
System.int LayerOverride {get; set;}
```

C++/CLI

```
property System.int LayerOverride {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Properties that have been overridden or should be overridden as defined in [swLayerOverride_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.LayerOverride](#).

Remarks

Layers are supported only in drawing documents.

In drawing documents, sketch points can be created on a layer that has specific visual properties. By default, the sketch point inherits the visual properties defined by the layer. However, for a specific sketch point, these visual properties can be overridden.

When the sketch point is not on any layer, the value this property returns is undefined. You can use the [ISketchPoint::Layer](#) property to determine the name of the layer, if any, that this sketch point is on. If an empty string is returned by the ISketchPoint::Layer property, then this property is not used.

Sketch points currently support only the color visual property. Therefore, the only bit value that is currently used by this function is 0x1, which indicates a color override. If this property returns 0x1, this indicates that the color of this sketch point has been specifically set by the end-user and may not match the default color value associated with this sketch point's layer. If you pass 0 to this property, then you are indicating that the color should be reset to use the default color associated with this item's layer.

Only set this property to reset specific visual properties back to the default visual properties of the owning layer. If you want to change or set specific values for the visual property for this item, use [ISketchPoint::Color](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Select Method (ISketchPoint)

Obsolete. Superseded by [ISketchPoint::Select4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select( _  
    ByVal AppendFlag As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim AppendFlag As System.Boolean  
Dim value As System.Boolean  
  
value = instance.Select(AppendFlag)
```

C#

```
System.bool Select(  
    System.bool AppendFlag  
)
```

C++/CLI

```
System.bool Select(  
& System.bool AppendFlag  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppendFlag

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Select](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Select2 Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Select2 Method (ISketchPoint)

Obsolete. Superseded by [ISketchPoint::Select4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select2( _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim Append As System.Boolean  
Dim Mark As System.Integer  
Dim value As System.Boolean  
  
value = instance.Select2(Append, Mark)
```

C#

```
System.bool Select2(  
    System.bool Append,  
    System.int Mark  
)
```

C++/CLI

```
System.bool Select2(  
& System.bool Append,  
& System.int Mark  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

Mark

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Select2](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select3 Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Select3 Method (ISketchPoint)

Obsolete. Superseded by [ISketchPoint::Select4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function Select3( _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer, _  
    ByVal Callout As Callout _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim Append As System.Boolean  
Dim Mark As System.Integer  
Dim Callout As Callout  
Dim value As System.Boolean
```

```
value = instance.Select3(Append, Mark, Callout)
```

C#

```
System.bool Select3(  
    System.bool Append,  
    System.int Mark,  
    Callout Callout  
)
```

C++/CLI

```
System.bool Select3(
& System.bool Append,
& System.int Mark,
& Callout^ Callout
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

Mark

Callout

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Select3](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Select4 Method (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Select4 Method (ISketchPoint)

Selects the sketch point and marks it.

.NET Syntax

Visual Basic (Declaration)

```
Function Select4( _
    ByVal Append As System.Boolean, _
    ByVal Data As SelectData _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim Append As System.Boolean
Dim Data As SelectData
Dim value As System.Boolean

value = instance.Select4(Append, Data)
```

C#

```
System.bool Select4(
    System.bool Append,
    SelectData Data
)
```

C++/CLI

```
System.bool Select4(
& System.bool Append,
& SelectData^ Data
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

True to append this selection to the selection list, false to replace the selection list with this selection

Data
[ISelectData](#) object

Return Value

True if the sketch point is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Select4](#).

Example

[Create Plane Thru 3 Points In-context \(VBA\)](#)
[Get Reference Point Data \(VBA\)](#)
[Get Sketch Points in Wizard Hole \(VBA\)](#)
[Insert Coordinate System at Center of Mass \(VBA\)](#)
[Insert DXF File and Add Dimension \(VBA\)](#)
[Get Sketch Point's Selection Mark \(C#\)](#)
[Get Sketch Point's Selection Mark \(VB.NET\)](#)
[Get Sketch Point's Selection Mark \(VBA\)](#)
[Get Sketch Relations \(VBA\)](#)
[Get Sketch Relations \(VB.NET\)](#)
[Get Sketch Relations \(C#\)](#)

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::Deselect Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SelectByMark Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : SelectByMark Method (ISketchPoint)

Obsolete. Superseded by [ISketchPoint::Select4](#).

.NET Syntax

Visual Basic (Declaration)

```
Function SelectByMark( _  
    ByVal AppendFlag As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim AppendFlag As System.Boolean  
Dim Mark As System.Integer  
Dim value As System.Boolean
```

```
value = instance.SelectByMark(AppendFlag, Mark)
```

C#

```
System.bool SelectByMark(  
    System.bool AppendFlag,  
    System.int Mark  
)
```

C++/CLI

```
System.bool SelectByMark(  
    & System.bool AppendFlag,
```

```
& System.int Mark
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppendFlag

Mark

Visual Basic for Applications (VBA) Syntax

See [SketchPoint.SelectByMark](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetCoords Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : SetCoords Method (ISketchPoint)

Sets the location of this sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCoords( _
    ByVal Xx As System.Double, _
    ByVal Yy As System.Double, _
    ByVal Zz As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim Xx As System.Double
Dim Yy As System.Double
Dim Zz As System.Double
Dim value As System.Boolean
```

```
value = instance.SetCoords(Xx, Yy, Zz)
```

C#

```
System.bool SetCoords(
    System.double Xx,
    System.double Yy,
    System.double Zz
)
```

C++/CLI

```
System.bool SetCoords(
    & System.double Xx,
    & System.double Yy,
    & System.double Zz
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Xx

X component of the direction vector

Yy

Y component of the direction vector

Zz
Z component of the direction vector

Return Value

True if the coordinates are successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::SetCoords](#).

Remarks

When setting new coordinate values to a sketch point, this method adheres to any constraints that are active in the sketch. If the sketch point is an end point of a sketch line that is constrained to be horizontal, then the values are adjusted according to that constraint.

To set the coordinates of a sketch point on a spline, the location of the sketch point must already exist on the spline. You cannot use this method to create a new sketch point that is not on the spline.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::GetCoords Method](#)
[ISketchPoint::X Property](#)
[ISketchPoint::Y Property](#)
[ISketchPoint::Z Property](#)

Availability

SOLIDWORKS 2000 FCS, Revision Number 8.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetFramePointTangent Method (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : SetFramePointTangent Method (ISketchPoint)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

```
Function SetFramePointTangent( _  
    ByVal ToVector As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim ToVector As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetFramePointTangent(ToVector)
```

C#

```
System.bool SetFramePointTangent(  
    System.object ToVector  
)
```

C++/CLI

```
System.bool SetFramePointTangent(  
& System.Object^ ToVector  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::SetFramePointTangent](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Status Property (ISketchPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Status Property (ISketchPoint)

Gets or sets the status of the associated relation for this sketch point.

.NET Syntax

Visual Basic (Declaration)

Property Status As System.Integer

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Integer
```

```
instance.Status = value
```

```
value = instance.Status
```

C#

```
System.int Status {get; set;}
```

C++/CLI

```
property System.int Status {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Sketch point relation status as defined in [swConstrainedStatus_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Status](#).

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

Type Property (ISketchPoint)

[See Also](#) [Example](#)



Gets or sets the type of sketch point.

.NET Syntax

Visual Basic (Declaration)

Property Type As System.Integer

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Integer
```

```
instance.Type = value
```

```
value = instance.Type
```

C#

```
System.int Type {get; set;}
```

C++/CLI

```
property System.int Type {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of sketch point as defined in [swSketchPointType_e](#)

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Type](#).

Example

[Get Persistent Identifiers and Types for Sketch Points \(VBA\)](#)
[Get Persistent Identifiers and Types for Sketch Points \(VB.NET\)](#)
[Get Persistent Identifiers and Types for Sketch Points \(C#\)](#)

Remarks

Sketch slot information is independent of the sketch point type.

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

X Property (ISketchPoint)

[See Also](#) [Example](#)



Gets or sets the x coordinate of the sketch point.

.NET Syntax

Visual Basic (Declaration)

Property X As System.Double

Visual Basic (Usage)



```
Dim instance As ISketchPoint
Dim value As System.Double
```

```
instance.X = value
```

```
value = instance.X
```

C#

```
System.double X {get; set;}
```

C++/CLI

```
property System.double X {
    System.double get();
    void set ( & System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

x coordinate of the sketch point location

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::X](#).

Example

[Get All Elements of Sketch \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Get Sketch Points \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::Y Property](#)
[ISketchPoint::Z Property](#)
[ISketchPoint::SetCoords Method](#)
[ISketchPoint::GetCoords Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

Y Property (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Y Property (ISketchPoint)

Gets or sets the y coordinate of the sketch point.

.NET Syntax

Visual Basic (Declaration)

Property Y As System.Double

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Double
```

```
instance.Y = value
```

```
value = instance.Y
```

C#

```
System.double Y {get; set;}
```

C++/CLI

```
property System.double Y {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

y coordinate of the point location

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Y](#).

Example

[Get All Elements of Sketch \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Get Sketch Points \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::X Property](#)
[ISketchPoint::Z Property](#)
[ISketchPoint::SetCoords Method](#)
[ISketchPoint::GetCoords Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments](#) on this topic.

Z Property (ISketchPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchPoint Interface](#) : Z Property (ISketchPoint)

Gets or sets the z coordinate of the sketch point.

.NET Syntax

Visual Basic (Declaration)

```
Property Z As System.Double
```

Visual Basic (Usage)



```
Dim instance As ISketchPoint  
Dim value As System.Double
```

```
instance.Z = value
```

```
value = instance.Z
```

C#

```
System.double Z {get; set;}
```

C++/CLI

```
property System.double Z {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

z coordinate of the point location

Visual Basic for Applications (VBA) Syntax

See [SketchPoint::Z](#).

Example

[Get All Elements of Sketch \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Get Spline's Parameters \(C#\)](#)
[Get Spline's Parameters \(VB.NET\)](#)
[Get Spline's Parameters \(VBA\)](#)

See Also

[ISketchPoint Interface](#)
[ISketchPoint Members](#)
[ISketchPoint::X Property](#)
[ISketchPoint::Y Property](#)
[ISketchPoint::SetCoords Method](#)
[ISketchPoint::GetCoords Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

ISketchRegion (Type)

ISketchRegion Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRegion Interface

Provides access to sketch regions.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface ISketchRegion
```

Visual Basic (Usage)

```
Dim instance As ISketchRegion
```

C#

```
public interface ISketchRegion
```

C++/CLI

```
public interface class ISketchRegion
```

Visual Basic for Applications (VBA) Syntax

See [SketchRegion](#).

Example

[Determine If Sketch Contour or Sketch Region \(VBA\)](#)
[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

Remarks

Sketch regions are white space on a sketch bordered by sketch entities or portions of sketch entities basically forming a face.
A centerline within a sketch creates sketch regions inside that sketch.

Accessors

[IExtrudeFeatureData2::Contours](#)
[IExtrudeFeatureData2::IGetContours](#)
[IRevolveFeatureData2::Contours](#)
[IRevolveFeatureData2::IGetContours](#)
[ISketch::GetSketchRegions](#) and [ISketch::IGetSketchRegions](#)
[ISplitLineFeatureData::Contours](#)
[ISplitLineFeatureData::IGetContours](#)

Access Diagram

[SketchRegion](#)

See Also

[ISketchRegion Members](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISketchRegion Interface Members

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks Namespace](#) : ISketchRegion Interface

The following tables list the members exposed by [ISketchRegion](#).

Public Properties

Name	Description
 Sketch	Gets the sketch for this sketch region.

[Top](#)

Public Methods

Name	Description
 Deselect	Deselects the sketch region.
 GetEdges	Gets the edges on this sketch region.
 GetEdgesCount	Gets the number of edges for this sketch region.
 GetFirstLoop	Gets the first loop in this sketch region.
 IGetEdges	Gets the edges on this sketch region.
 Select	Obsolete. Superseded by ISketchRegion::Select2
 Select2	Selects the sketch region and marks it.

[Top](#)

See Also

[ISketchRegion Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISketchRegion Interface Methods

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : ISketchRegion Interface

For a list of all members of this type, see [ISketchRegion members](#).

Public Methods

Name	Description
 Deselect	Deselects the sketch region.
 GetEdges	Gets the edges on this sketch region.
 GetEdgesCount	Gets the number of edges for this sketch region.
 GetFirstLoop	Gets the first loop in this sketch region.
 IGetEdges	Gets the edges on this sketch region.
 Select	Obsolete. Superseded by ISketchRegion::Select2
 Select2	Selects the sketch region and marks it.

[Top](#)

See Also

[ISketchRegion Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

ISketchRegion Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISketchRegion Interface

For a list of all members of this type, see [ISketchRegion members](#).

Public Properties

Name	Description
 Sketch	Gets the sketch for this sketch region.

[Top](#)

See Also

[ISketchRegion Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

DeSelect Method (ISketchRegion)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : DeSelect Method (ISketchRegion)

Deselects the sketch region.

.NET Syntax

Visual Basic (Declaration)

```
Function DeSelect() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim value As System.Boolean  
  
value = instance.DeSelect()
```

C#

```
System.bool DeSelect()
```

C++/CLI

```
System.bool DeSelect();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the sketch region is deselected, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::DeSelect](#).

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)
[ISketchRegion::Select2 Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEdges Method (ISketchRegion)

[See Also](#)



Gets the edges on this sketch region.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdges() As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim value As System.Object
```

```
value = instance.GetEdges()
```

C#

```
System.object GetEdges()
```

C++/CLI

```
System.Object^ GetEdges();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Array of [edges](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::GetEdges](#).

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)
[ISketchRegion::GetEdgesCount Method](#)
[ISketchRegion::!GetEdges Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

GetEdgesCount Method (ISketchRegion)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [ISketchRegion Interface](#) : GetEdgesCount Method (ISketchRegion)

Gets the number of edges for this sketch region.

.NET Syntax

Visual Basic (Declaration)

```
Function GetEdgesCount() As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim value As System.Integer
```

```
value = instance.GetEdgesCount()
```

C#

```
System.int GetEdgesCount()
```

C++/CLI

```
System.int GetEdgesCount();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Number of edges

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::GetEdgesCount](#).

Remarks

Call this method before calling [ISketchRegion::IGetEdges](#) to get the size of the array for that method.

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)
[ISketchRegion::GetEdges Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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GetFirstLoop Method (ISketchRegion)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : GetFirstLoop Method (ISketchRegion)

Gets the first loop in this sketch region.

.NET Syntax

Visual Basic (Declaration)

```
Function GetFirstLoop() As Loop2
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim value As Loop2  
  
value = instance.GetFirstLoop()
```

C#

```
Loop2 GetFirstLoop()
```

C++/CLI

```
Loop2^ GetFirstLoop();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

First [loop](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::GetFirstLoop](#).

Example

[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)

Availability

SOLIDWORKS 2009 FCS, Release Number 17.0

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IGetEdges Method (ISketchRegion)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : IGetEdges Method (ISketchRegion)

Gets the edges on this sketch region.

.NET Syntax

Visual Basic (Declaration)

```
Function IGetEdges( _  
    ByVal Count As System.Integer _  
) As Edge
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim Count As System.Integer  
Dim value As Edge
```

```
value = instance.IGetEdges(Count)
```

C#

```
Edge IGetEdges(  
    System.int Count  
)
```

C++/CLI

```
Edge^ IGetEdges(  
& System.int Count  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Count
Number of edges

Return Value

- in-process, unmanaged C++: Pointer to an array of [edges](#)
 - VBA, VB.NET, C#, and C++/CLI: Not supported
- See [In-process Methods](#) for details about this type of method.

Remarks

Before calling this method, call [ISketchRegion::GetEdgesCount](#) to get the value for Count.

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)
[ISketchRegion::GetEdges Method](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

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Select Method (ISketchRegion)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : Select Method (ISketchRegion)

Obsolete. Superseded by [ISketchRegion::Select2](#)

.NET Syntax

Visual Basic (Declaration)

```
Function Select( _  
    ByVal Append As System.Boolean, _  
    ByVal Mark As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim Append As System.Boolean  
Dim Mark As System.Integer  
Dim value As System.Boolean  
  
value = instance.Select(Append, Mark)
```

C#

```
System.bool Select(  
    System.bool Append,  
    System.int Mark  
)
```

C++/CLI

```
System.bool Select(  
& System.bool Append,  
& System.int Mark  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

Mark

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::Select](#).

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Select2 Method (ISketchRegion)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : Select2 Method (ISketchRegion)

Selects the sketch region and marks it.

VB.NET Syntax

Visual Basic (Declaration)

```
Function Select2( _  
    ByVal Append As System.Boolean, _  
    ByVal Data As SelectData _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim Append As System.Boolean  
Dim Data As SelectData  
Dim value As System.Boolean  
  
value = instance.Select2(Append, Data)
```

C#

```
System.bool Select2(  
    System.bool Append,  
    SelectData Data  
)
```

C++/CLI

```
System.bool Select2(  
& System.bool Append,  
& SelectData^ Data  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Append

True to append this selection to the selection list, false to replace the selection list with this selection

Data

[ISelectData](#) object

Return Value

True if the sketch region is selected, false if not

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::Select2](#).

Example

[Get Sketch Regions \(VBA\)](#)
[Get Sketch Regions \(VB.NET\)](#)
[Get Sketch Regions \(C#\)](#)

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)
[ISketchRegion::DeSelect Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

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Sketch Property (ISketchRegion)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISketchRegion Interface](#) : Sketch Property (ISketchRegion)

Gets the sketch for this sketch region.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property Sketch As Sketch

Visual Basic (Usage)



```
Dim instance As ISketchRegion  
Dim value As Sketch
```

```
value = instance.Sketch
```

C#

```
Sketch Sketch {get;}
```

C++/CLI

```
property Sketch^ Sketch {  
    Sketch^ get();  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Sketch](#)

Visual Basic for Applications (VBA) Syntax

See [SketchRegion::Sketch](#).

See Also

[ISketchRegion Interface](#)
[ISketchRegion Members](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0